

# Predicting the onset of lithium plating by the full-cell voltage: A pseudo-P curve approach

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## ABSTRACT

Fast-charging lithium-ion batteries (LIBs) may cause the formation of lithium metal deposits, known as lithium plating, on the anode, which not only reduces battery capacity but also increases the risk of internal short circuits, posing safety hazards. However, determining the onset of lithium plating solely from full-cell data of commercial LIBs, without disassembly, remains a challenge. To address this, we introduce a novel experimental method that generates a pseudo-P curve. This curve shares the same profile as the plating current curve (which can be simulated but not directly measured), varying only by a specific vertical scaling factor. It is found that the intersection of the two curves pinpoints the onset of lithium deposition. Constructing the pseudo-P curve involves identifying the end time of lithium stripping and the corresponding net discharge capacity by varying the charging cutoff voltages. To validate the method's ability to correctly detect lithium plating onset, direct electrochemical simulations are conducted, showing a prediction error within 3.44 % of total capacity. The correctness of this method is further supported by indirect experimental validation, based on the presence or absence of stripping signals when the charging duration is shorter or longer than the lithium plating onset point. Using the plating onset as a turning point, this method effectively guides charging strategies through current reduction at those turning moments. The proposed step-charging strategy fully prevents lithium plating while achieving a Coulomb efficiency of 99.7 % under the 2 C step-charging scenario.

## 1. Introduction

As modern technology rapidly evolves, the serious consequences of global warming and the growing societal focus on renewable energy have made electric vehicles (EVs) increasingly popular among consumers. Lithium-ion batteries (LIBs) are the primary energy storage technology for EVs, chosen for their impressive energy density, excellent voltage performance, long lifespan, and low self-discharge rates. However, LIBs encounter significant issues, particularly lithium plating, which tends to develop during fast charging [1], at low temperatures [2], when highly charged [3], and as the batteries age over time. These conditions can increase the local ion flux, slow lithium-ion diffusion, and reduce the availability of insertion sites in the anode, which causes intercalation kinetics to become insufficient to accommodate incoming ion flux and thereby causes lithium deposition on the anode surface [4]. The accumulation of lithium metal on the anode can reduce capacity, raise the risk of internal short circuits, and compromise safety [5–7]. Therefore, detecting and preventing lithium plating has become a crucial area of research not only in battery technology but also in the

development and safety of electric vehicles [8].

Several methods have been proposed to identify lithium plating, encompassing both invasive and non-invasive detection techniques [9, 10]. Among the former methods, the reference electrode approach [11, 12] allows for direct measurement of anode potential, facilitating accurate detection of lithium plating onset. However, due to its requirement for additional electrodes within the battery, this technique is predominantly limited to laboratory environments, rendering it unsuitable and potentially hazardous for deployment in commercial applications. Other invasive techniques, such as scanning electron microscopy (SEM) [13,14], provide detailed insights into battery structures but are costly and demand extensive sample preparation, restricting their use to laboratory environments. Mass spectrometry titration (MST) [15], which analyzes components of the solid electrolyte interface (SEI), is another method; however, it is constrained by the need for high-precision calibration.

Various non-invasive techniques have also been developed. The most commonly used detecting methods include electrochemical impedance spectroscopy (EIS) [16–18], incremental capacity (IC) analysis [19,20],

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the relaxation voltage (RV) analysis [21,22], differential voltage (DV) analysis [23,24], differential pressure sensing (DPS) [25], and Coulombic efficiency (CE) monitoring [26]. Although EIS offers valuable insights by tracking the evolution of charge-transfer resistance, it typically requires extended relaxation times and costly equipment, which restricts its real-time applicability. Advancements such as dynamic EIS have mitigated these limitations by permitting continuous frequency-domain analysis during the charging phase [18,27]. IC, RV, and DV analysis can indirectly estimate lithium plating based on the voltage-time curves; however, the accuracy for pinpointing the onset of plating has not yet been fully confirmed. This is because the voltage curve changes only slightly when the deposited lithium amount is small. DPS is able to detect the subtle internal pressure changes associated with electroplating, but is highly susceptible to external interference and demands precise calibration. CE analysis is another non-invasive method that detects lithium plating by tracking inefficiencies during charge/discharge cycles. While simple and accessible, its sensitivity is limited, particularly in distinguishing plating from other aging effects.

To analyze and forecast lithium plating, modeling and data-driven approaches are becoming the booming trend [28]. Physically-based electrochemical models (e.g., a pseudo-two-dimensional (P2D) model [29,30], a multi-scale particle-level deposition-stripping model [31,32], a mathematical overcharge model [33], and a volume expansion-based detection model [34]) provide a theoretical framework to simulate lithium deposition under various charging conditions. While these models can accurately estimate the plating overpotential and lithium distribution, they require extensive parameterization of the cell's internal state and considerable computational resources. On the other hand, machine learning (ML) models [35] use large datasets such as voltage, current, and temperature to predict electroplating risks. While ML approaches hold promise for enabling point-of-care diagnostics, their predictive accuracy depends heavily on the quality and diversity of the training data and they often lack interpretability, limiting their acceptance in safety-critical battery management systems. Overall, these model-related methods are valuable tools for understanding and tackling lithium plating, though they are currently more applicable for research and development than for direct commercial applications.

To address the limitations and challenges mentioned above, this study proposes a pseudo-P curve method that integrates direct measurements from commercial LIBs with electrochemical simulations to determine the onset of lithium plating. The theoretical foundation of this method is thoroughly explained, and its validity is confirmed through simulations and full-cell experiments. This method also supports multi-stage constant current (MSSC) charging strategies, enabling higher charging rates while completely eliminating lithium plating, as validated by model-based [36], data-driven [37], and experimental studies [38,39]. Table 1 summarizes the strengths and limitations of commonly

used lithium plating detection methods with respect to five key criteria: battery disassembly, exact plating onset prediction, physical interpretability, compatibility with charging strategies, and suitability for real-time battery management systems (BMS). Comparisons between our method and several other non-invasive approaches will be presented in the subsequent discussion section. While most conventional techniques fail to satisfy all criteria simultaneously, the proposed method meets every benchmark, highlighting its potential for practical deployment in embedded BMS platforms.

The main contributions of this study can be summarized as follows:

- (1) Generation of the proposed pseudo-P curve using only full-cell signals: Experimental data from commercial batteries are utilized to construct this curve, from which the onset of lithium plating is determined.
- (2) Analysis of the relationship between the plating current curve (P curve) and the pseudo-P curve: Electrochemical simulations are performed to ensure the consistent ratio between the plating amount during charging and the net discharge capacity during discharging.
- (3) Application of step-charging strategies: The onset of lithium plating is used as the transition point to switch the current magnitude, leading to the proposal and evaluation of step-charging protocols.

The structure of this paper is organized as follows. Section 2 introduces the methodology for constructing the pseudo-P curve and its ability to predict the onset of lithium plating. Section 3 outlines the details of the experiment and data processing. Section 4 gives the experimental results, validates the prediction of plating onset, discusses step-charging strategies based on the onset of plating, further elaborates on the theoretical foundation of the method, and presents comparisons with other techniques. Section 5 gives the conclusion of this study.

## 2. Methodology

The flowchart for detecting lithium plating and optimizing charging strategies is shown in Fig. 1. The process begins with theoretical considerations—starting from overpotential-driven plating, progressing to plating current, and ultimately aiming to estimate the plating amount through voltage signals observed during the charge/discharge cycle. Electrochemical simulations are then conducted to support the theoretical framework and validate the proposed pseudo-P curve. At the next stage, the experimental voltage signals are employed to construct the pseudo-P curves at three different charging C-rates, enabling estimation of the onset of lithium plating. After detecting plating, we use this information to develop step-charging protocols. By implementing stepwise reductions in C-rate at identified lithium plating onsets, the proposed

**Table 1**  
Methods for detecting lithium plating with various pros and cons.

| Method              | Ref.      | Feature                           | No need of disassembly | Exact onset prediction | Physical interpretability | Charging strategy | Suitable for BMS |
|---------------------|-----------|-----------------------------------|------------------------|------------------------|---------------------------|-------------------|------------------|
| Reference electrode | [11]      | Anode potential                   | ×                      | ×                      | ✓                         | ✓                 | ×                |
| SEM                 | [13]      | SEM images                        | ×                      | ×                      | ×                         | ×                 | ×                |
| MST                 | [15]      | Correlation analysis              | ×                      | ×                      | ✓                         | ×                 | ×                |
| EIS                 | [16]      | $C_{dl}$ , $R_{CT}$ , $\tau_{CT}$ | ✓                      | ×                      | ✓                         | ✓                 | ×                |
| IC                  | [19]      | IC curve peaks                    | ✓                      | ×                      | ✓                         | ✓                 | ✓                |
| RV                  | [21]      | dV/dt and current plateau         | ✓                      | ×                      | ✓                         | ×                 | ✓                |
| DV                  | [23]      | dV/dQ valley                      | ✓                      | ×                      | ✓                         | ×                 | ✓                |
| DPS                 | [25]      | dP/dQ fluctuation                 | ✓                      | ×                      | ✓                         | ✓                 | ×                |
| CE                  | [26]      | Loss deviation                    | ✓                      | ×                      | ✓                         | ×                 | ✓                |
| P2D                 | [29]      | $\eta$ or $C_{max}$               | ✓                      | ✓                      | ✓                         | ✓                 | ×                |
| ML                  | [35]      | V, T, I                           | ▲                      | ✓                      | ×                         | ✓                 | △                |
| Pseudo-P curve      | This work | dV/dQ valley, net discharge       | ✓                      | ✓                      | ✓                         | ✓                 | ✓                |

Notes: ✓ represents “yes”; × denotes “no”; ▲ means both “yes” and “no” depending on cases; △ relies on model complexity and system constraints.

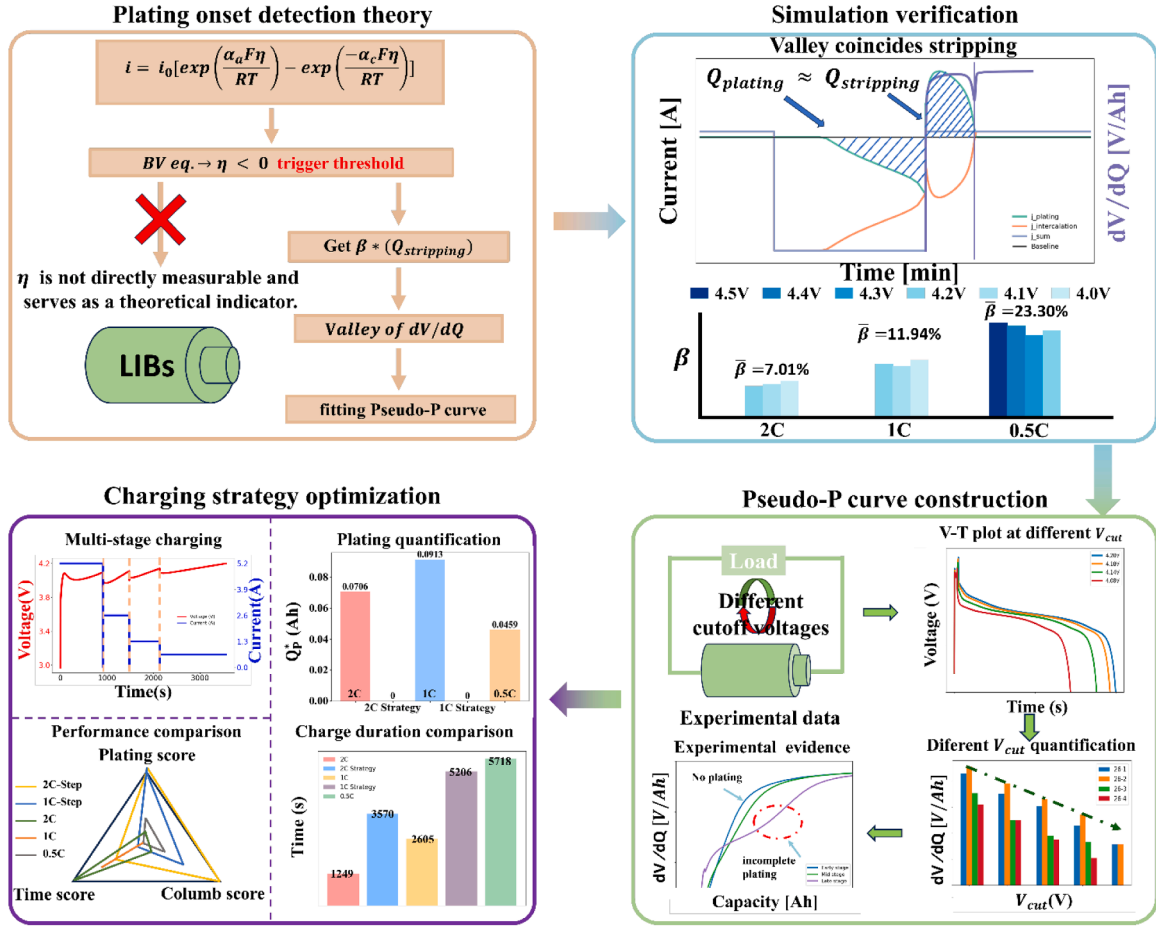


Fig. 1. Flowchart for detecting lithium plating and recommending charging strategy.

strategy effectively minimizes lithium plating while maintaining high Coulombic efficiency.

### 2.1. Theoretical thinking

When the plating overpotential—defined as the electrochemical potential difference between the anode surface and the electrolyte relative to the  $\text{Li}^+/\text{Li}$  redox potential—becomes negative (below 0 V versus  $\text{Li}/\text{Li}^+$ ), conditions become thermodynamically favorable for lithium metal deposition [40,41]. Therefore, the overpotential serves as an indicator of the onset of lithium deposition. A key challenge is whether this potential can be monitored in commercial LIBs without disassembling the cell to insert a reference electrode. To investigate this, we begin with the classical Butler–Volmer (BV) kinetics, which governs the charge transfer rate of electrode reactions. Under charging conditions, the total local reaction current density at the anode is the sum of two components: the lithium-ion intercalation current density  $j_1$ , and the lithium metal plating current density  $j_2$  [40–43]:

$$j = j_1 + j_2 \quad (1)$$

Both components can be expressed using the modified BV equations:

$$j_1 = j_{0,1} \left[ \exp\left(\frac{\alpha_{a,1} F \eta_1}{RT}\right) - \exp\left(\frac{-\alpha_{c,1} F \eta_1}{RT}\right) \right] \quad (2)$$

where  $j_{0,1}$  is the exchange current density for the intercalation reaction,  $F$  is Faraday constant,  $\alpha_{a,1}$  and  $\alpha_{c,1}$  are the anodic and cathodic transfer coefficients (typically both 0.5), and  $\eta_1$  is the overpotential for intercalation.

$$j_2 = j_{0,2} \left[ \exp\left(\frac{\alpha_{a,2} F \eta_2}{RT}\right) - \exp\left(\frac{-\alpha_{c,2} F \eta_2}{RT}\right) \right], \text{ for charging } (\eta_2 < 0) \quad (3)$$

Here,  $j_{0,2}$  is the exchange current density for the lithium plating reaction, and  $\alpha_{a,2}$  and  $\alpha_{c,2}$  are typically unequal (e.g., 0.3 and 0.7, respectively), reflecting the asymmetry of the plating mechanism. It should be noted that, during the transition from charging to discharging, the plating current density  $j_2$  is interpreted as the stripping current density:

$$j_2 = j_{0,3} \left[ \exp\left(\frac{\alpha_{a,2} F \eta_2}{RT}\right) - \exp\left(\frac{-\alpha_{c,2} F \eta_2}{RT}\right) \right], \text{ for discharging } (\eta_2 > 0) \quad (4)$$

where  $j_{0,3}$  is the exchange current density for the lithium stripping reaction [44].

During the charging process, lithium plating initiates as soon as the plating overpotential drops below zero ( $\eta_2 < 0$ ), corresponding to the onset of plating current (see Eq. (3)) [42,43]. Accordingly, identifying the zero plating overpotential point is equivalent to pinpointing the exact moment when plating current first starts to flow, effectively shifting the above challenge to monitoring the plating current. The following two-step idea is now proposed to extract plating current data from experiments conducted on commercial LIBs.

**STEP I:** Since reversible lithium plating can be completely stripped during discharge, it appears possible to determine the total amount of plated lithium by analyzing the discharge voltage profile of a full cell. Voltage differential ( $dV/dQ$ ) analysis, derived from processing the voltage signal, is recognized as a highly sensitive, non-

destructive technique commonly employed in LIB research, especially for detecting lithium deposition and stripping processes [45]. The valleys in the  $dV/dQ$  curve reveal the changes in voltage rate as lithium ions enter or exit the electrode surface [46,47]. Consequently, these distinctive valley points offer crucial information for the onset and completion of lithium plating and stripping, helping to identify the timing of these processes. Through examination of these valleys, it may be expected that the amount of reversible lithium plating occurring over a specific time interval can be estimated.

STEP II: Given the total amount of lithium plating, we can calculate the average plating current between any two timepoints, which serves as an indicator for determining when lithium plating begins to occur. However, the feasibility of our proposed idea—quantifying reversible lithium plating amounts and estimating average plating currents using discharge measurement data—requires careful validation, which will be comprehensively examined in the following section.

## 2.2. Electrochemical simulation of plating current

To investigate the possible relationship between the charge/discharge curves and the amount of lithium plating, we conduct electrochemical simulations utilizing the open-source Python Battery Mathematical Modeling (PyBaMM) package available on GitHub [48]. Thanks to its flexibility and computational efficiency, this package has recently gained widespread use in battery modeling [49,50]. The parameter dataset for the LG INR M50T ( $\text{LiNiMnCoO}_2$ , NMC) cell is utilized in the modeling [51], along with parameters related to lithium plating kinetics [52,53].

The simulations consider lithium plating as a competing side reaction that follows a thermodynamic criterion. Specifically, plating only begins when the local overpotential against the  $\text{Li}^+/\text{Li}$  redox potential drops below zero ( $\eta_2 < 0$ ), as shown in Eq. (3). The corresponding plating and stripping currents are calculated using overpotential-dependent BV expressions, allowing a continuous transition between deposition and dissolution without the need to manually switch reaction modes. Aging-related effects such as SEI layer growth, pore clogging, or lithium trapping are not taken into account in the present simulations, which may cause slight deviations in the actual onset of lithium plating in real LIBs. Assuming fully reversible lithium plating, the simulations are carried out under consistent conditions at 0 °C, beginning with a 0.05 C constant-current (CC) discharge to 2.5 V, followed by 1 C charging to three different cut-off voltages (4.00 V, 4.10 V, and 4.20 V), and ending with a second 0.05 C discharge to 2.5 V.

Our simulations clearly demonstrate the mechanism underlying the transition of lithium ions from intercalation to plating, followed by re-intercalation. As illustrated in Fig. 2, the 1 C charging and 0.05 C discharging processes can be divided into three stages. In the early stage of charging, lithium ions intercalate into the graphite anode. As charging proceeds and the intercalation capacity gradually saturates near the anode surface, lithium plating occurs during the later phase of charging. When discharge at a low C-rate begins, lithium ions from the plated lithium are first stripped into the electrolyte, then driven by

concentration gradients and electric potential differences to migrate back into the anode.

Fig. 3(a) displays the voltage evolution of a representative simulation cycle. The four key time points,  $T_0$ ,  $T_A$ ,  $T_B$ , and  $T_C$ , are specially highlighted, corresponding to the first transition between charge and discharge, the onset of lithium plating, the second transition between charge and discharge, and the end of lithium stripping, respectively. Fig. 3(b) presents the time evolutions of the currents associated with lithium plating (blue line), intercalation (orange line), and their combined total (green line). Note that, in the simulations, the current density in anode is transformed into current by multiplying the active area, which is evaluated by  $(3\varepsilon_s/R_p)LA$  with  $\varepsilon_s$ ,  $R_p$ ,  $L$ , and  $A$  being the volume fraction of solid particles, particle radius, electrode thickness, and cross sectional area, respectively [54].

Symbols (i) through (vi) signify the six current curves during the time intervals between  $T_A$  and  $T_B$ , and between  $T_B$  and  $T_C$ . To be explicit, Symbol (i) denotes the plating current curve (referred to as the P curve for simplicity), which emerges during the late stage of charging. This curve is rarely discussed due to the experimental challenges in measuring it, yet it can offer valuable dynamic details on lithium deposition in simulations. Symbol (ii) refers to the stripping current curve observed at the beginning of discharge, when lithium metal starts to dissolve back into the electrolyte. Symbol (iii) indicates the curve for the intercalation current during charging. This current initially equals the total current but gradually declines as plating commences, highlighting the trade-off between intercalation and plating needed to maintain a constant total current. Symbol (iv) represents the re-intercalation current curve in discharging phase, where stripped lithium ions migrate back into the graphite. Symbol (v) stands for the curve for the total current, which stays constant (1 C) throughout the process by the definition of CC charging. Symbol (vi) is the curve for the net discharge current across the time period between  $T_B$  and  $T_C$ , which is the sum of the stripping (ii) and re-intercalation (iv) currents, and it remains nearly constant (0.05 C) to maintain the balance of the total current.

Concerning the area analysis in Fig. 3(b), the blue area enclosed by the P curve between points  $T_A$  and  $T_B$  is equal in magnitude to the blue area under the blue line between  $T_B$  and  $T_C$ . This indicates that the total amount of lithium plating during charging is fully dissolved during discharging, assuming only reversible plating is considered. Moreover, during the discharge period from  $T_B$  and  $T_C$ , the difference between the absolute values of the areas enclosed by the blue and orange lines equals the area enclosed by the green line. This green, slashed-line region represents the net discharge capacity over the time interval from  $T_B$  to  $T_C$ , which can be directly obtained from experimental measurements.

Moreover, the simulations also reveal that the end point  $T_C$ , at which the stripping current vanishes, nearly aligns with the valley point of the differential voltage  $dV/dQ$  curve during the discharging phase. This assertion, validating the use of the  $dV/dQ$  valley as a reliable marker for the end of lithium stripping, is substantiated and depicted in Fig. 4(a), which presents three scenarios with charging cutoff voltages of 4.0 V, 4.1 V and 4.2 V. As the time points  $T_B$  and  $T_C$  can be experimentally determined, the net discharge capacity can be measured under different

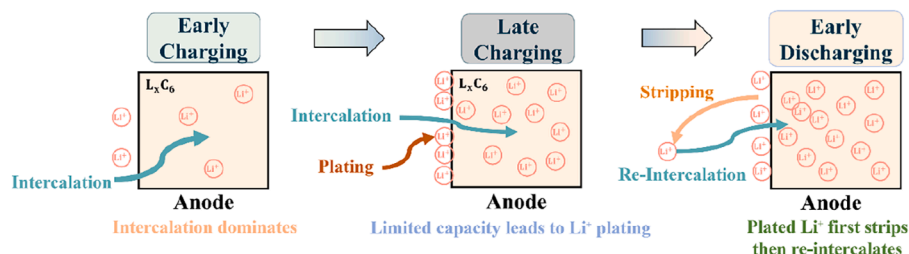


Fig. 2. Schematic of lithium-ion movement during early and late charging, as well as early discharging.

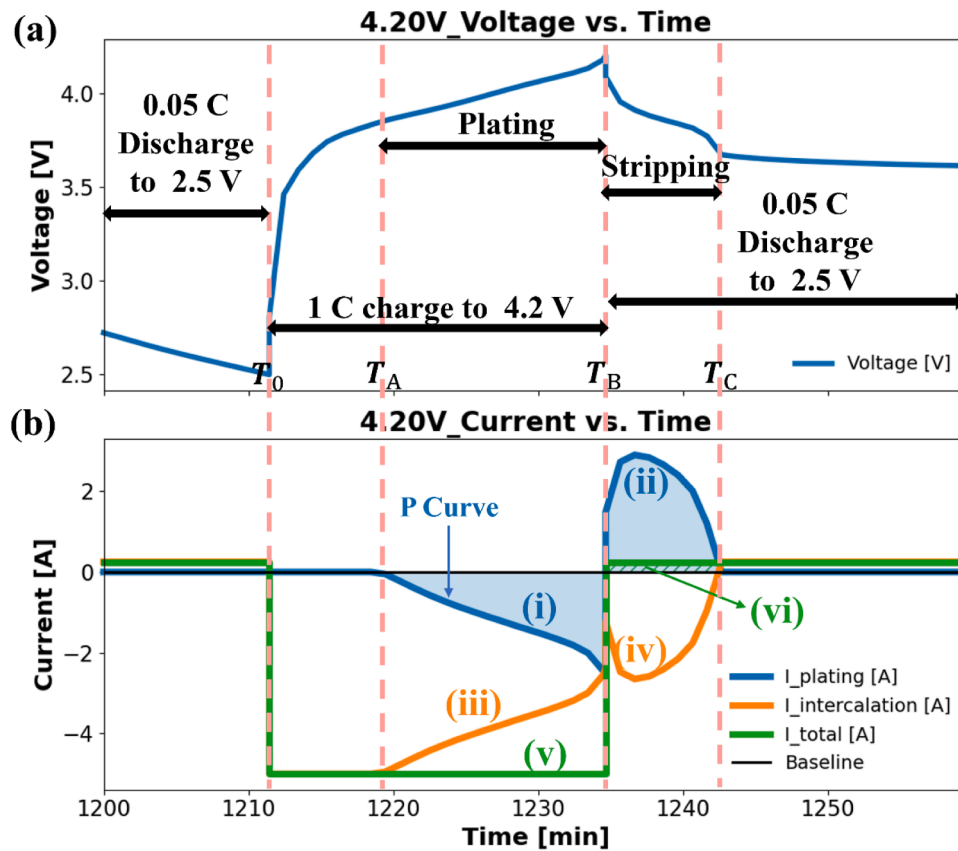


Fig. 3. (a) Voltage profile of a 0.05 C–1 C–0.05 C cycle. (b) Time dependent currents depicting lithium plating, intercalation, and their sum.

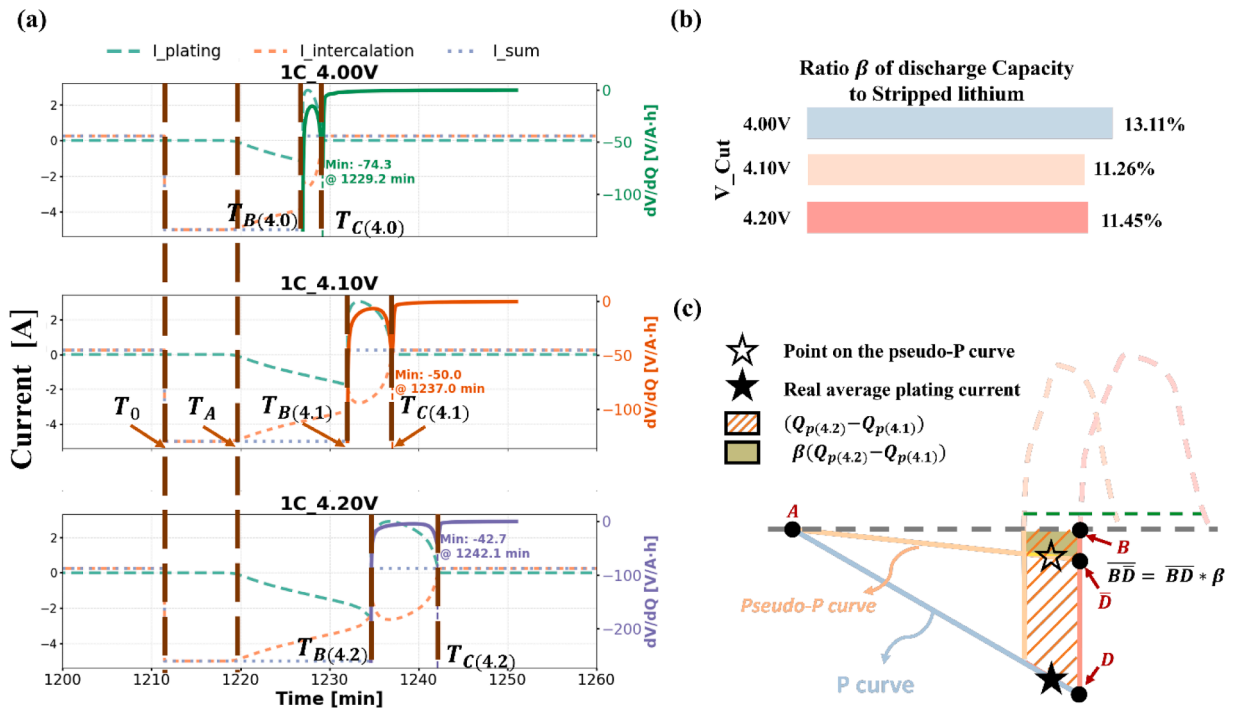


Fig. 4. (a) dV/dQ valley marks the end of lithium stripping. (b) Ratio  $\beta$  of discharge capacity to stripping lithium amount at three different charging cutoff voltages. (c) The P curve and its  $\beta$ -ratio-modified pseudo-P curve, intersecting at time  $T_A$  (point A).

charging cutoff voltages. Fig. 4(a) also illustrates that the P curves in the three scenarios follows the same trajectory. Specifically, at the same charging C-rate but with different cut-off voltages of 4.0 V, 4.1 V, and

4.2 V, the P curve begins at the same time point ( $T_A$ ) and follows the same path to reach different time points:  $T_B(4.0)$ ,  $T_B(4.1)$ , and  $T_B(4.2)$ . This property is essential for the subsequent analysis, as it allows the



determination of the starting time point  $T_A$  by examining the unique P curve.

The P curve could be constructed skillfully and strategically once at least a few points on it are identified. To clearly interpret the geometric significance of the P curve, we define three points A, B, and D in Fig. 4(c) as the point on the P curve at  $T_A$ , the P curve's projection onto the baseline at  $T_B$ , and the point on the P curve at  $T_B$ . The area enclosed by the points A, B, and D, which corresponds to the absolute value of the area between the P curve and the baseline within the interval between  $T_A$  and  $T_B$ , represents the amount of lithium plating  $Q_p$  (also the same as the lithium stripping between  $T_B$  and  $T_C$ , as also indicated in Fig. 3(b)). However, this area  $Q_p$  cannot be directly obtained from measurements, making it impossible to generate the points on the P curve and thus creating an obstacle to producing the P curve. To avoid this obstacle, a novel approach for determining the plating onset  $T_A$  is introduced here, using a proposed pseudo-P curve described in the following three steps. The three steps are (i) establishing the relationship between the net discharge capacity and the stripping amount, (ii) proposing a new curve proportional to the P curve, and (iii) generating the new curve from experimental data.

First, we evaluate the ratio (called  $\beta$  ratio) of net discharge capacity to the amount of stripped lithium during the discharge process between  $T_B$  and  $T_C$ . Here, the net discharge capacity—an experimentally measurable quantity—is represented by the slashed area ( $=I_{total}(T_C - T_B)$ ) under the green line in Fig. 3(b), whereas the amount of stripped lithium, which cannot be directly measured experimentally, corresponds to the area between the blue line and the baseline. As shown in Fig. 4(b), the  $\beta$  ratio consistently stays around 12 % in the simulations, nearly independent of the charging cutoff voltage. Further discussion of various  $\beta$  values will be provided in the subsequent section. Owing to the consistent  $\beta$  ratio, the amount of stripping (or plated) lithium  $Q_p$  can be inferred from the measurable net discharge capacity  $Q_p^*$ , where  $Q_p^*$  is defined as  $\beta Q_p$ .

Second, let us take the cutoff voltage of 4.2 V as an example and focus on the P curve between  $T_A$  and  $T_B$ . Instead of directly evaluating the lithium plating amount  $Q_p$ , we seek a region within  $T_A$  and  $T_B$  that has an area equal to  $Q_p^*$  ( $=\beta Q_p = I_{total}(T_C - T_B)$ ). This can be achieved by generating a new curve, called the pseudo-P curve, which resembles the P curve but is modified by the  $\beta$  ratio in the vertical direction. As shown in Fig. 4(c), the absolute value of the area between the pseudo-P curve and the baseline over the interval from  $T_A$  and  $T_B$ —enclosed by points A, B, and  $\bar{D}$ —represents the measurable discharge capacity  $Q_{p(4.2)}^*$ , where  $\bar{D}$  is the point on the pseudo-P curve at time  $T_B$ . In addition, the pseudo-P curve also passes through the same point A, which is exactly the target we aim to determine.

Third, to implement the generation of pseudo-P curve from the simulation perspective, several points on this curve must be determined, allowing its construction based on the measurable net discharge capacity. This can be readily achieved by using two cutoff voltages—such as 4.2 V and 4.1 V—as an illustrative example. It should be noted from Fig. 4(c) that the slashed area is  $(Q_{p(4.2)} - Q_{p(4.1)})$ , with its yellow portion corresponding to  $(Q_{p(4.2)}^* - Q_{p(4.1)}^*)$ . Thus, the points on the P curve and the pseudo-P curve at time  $(T_{B(4.2)} + T_{B(4.1)})/2$ , symbolled by “★” and “☆”, can be approximately located by evaluating  $(Q_{p(4.2)} - Q_{p(4.1)})/(T_{B(4.2)} - T_{B(4.1)})$  and  $(Q_{p(4.2)}^* - Q_{p(4.1)}^*)/(T_{B(4.2)} - T_{B(4.1)})$ , where  $Q_{p(4.2)}$  and  $Q_{p(4.1)}$  denotes the total amounts of lithium plating when the charging ends at 4.2 V and 4.1 V, respectively. Similarly, other points on the two curves can be determined using the same procedure.

In conclusion, the proposed pseudo-P curve offers two main advantages. (i) It closely resembles the P curve and shares a common intersection point at time  $T_A$ , which marks the onset of lithium plating. (ii) It can be determined using detectable full-cell data. Therefore, even without knowing the exact lithium plating amount and the  $\beta$  ratio, the

onset time of lithium plating  $T_A$  can still be inferred by extrapolating points on the curve—each derived from the measurement of  $T_B$ ,  $T_C$ , and net discharge capacity  $I_{total}(T_C - T_B)$  within the interval between  $T_B$  and  $T_C$ .

### 3. Experiment and data processing

Previous electrochemical studies indicated that lithium plating initiates during CC charging when the anode overpotential drops below 0 V (vs. Li/Li<sup>+</sup>). As CC charging continues, the plating current increases accordingly. The goal of this experiment is to determine the moment when the plating current starts using only full-cell measurement data, where the differences in the discharge capacity between any two selected data points are evaluated to derive the pseudo-P curve.

#### 3.1. Experimental setup

Four brand-new commercial Samsung 18,650–26 J cylindrical NMC LIBs, labeled 26–1, 26–2, 26–3, and 26–4, were tested using a Bio-Logic BCS-815 ultra-precision battery cycler and a Bio-Logic BH-1i commercial battery holder. All tests were conducted in a controlled temperature environment using an ABT-CH-R60 thermal chamber. The experimental procedures include activation, state of health (SOH) evaluation, and lithium plating tests, as illustrated in Fig. 5(a).

#### 3.2. Various tests

**Activation**—Before the experiment, batteries underwent an activation process to stabilize the solid electrolyte interphase (SEI). Each cell was charged using a CC protocol at 0.5 C up to 4.2 V, followed by a constant voltage (CV) phase until the current decreased below 0.05 C, and then discharged at a CC of 0.2 C down to 2.75 V. Additional five 1 C charge/discharge cycles were carried out to accelerate SEI stabilization and establish a consistent base capacity matching the nominal value.

**SOH measurement**—To consider the possible minor capacity loss from repeated cycling of a single battery at low temperatures, the SOH was evaluated following the activation process and after each plating test. Each SOH test involved a 0.5 C CC charge to 4.2 V, a CV charge with a cutoff current of 0.05 C, a 1-hour rest, and a 0.2 C discharge to 2.75 V. The resulting discharge capacity was normalized by the nominal capacity (2600 mAh) to define the SOH. To eliminate the SOH-induced bias in charging time (CT), a correction factor was applied to ensure consistency across tests

$$SOH_{avg} = \frac{(SOH_1 + \dots + SOH_n)}{n} \quad (5)$$

$$CT_{i,corr} = \frac{CT_i}{SOH_i} \times SOH_{avg} \quad (6)$$

where each test refers to a single charge/discharge cycle,  $SOH_i$  is the SOH value of test  $i$ ,  $SOH_{avg}$  is the average SOH across all test cases,  $CT_i$  represents the observed CC charging time, and  $CT_{i,corr}$  means the normalized charging time for use in plating current estimation.  $n$  denotes the number of distinct charge/discharge test cases conducted on the same battery under varying cut-off voltages, which is 4 for 2 C charging and 5 for 1 C charging.

**Lithium plating test**—Two plating test protocols were conducted at 0 °C. All cells were initially discharged to 2.75 V at 0.05 C to standardize initial conditions, with voltage, current, capacity, and time recorded every five seconds during subsequent charging and discharging. The procedure of this test is outlined in Fig. 5(b).

(1) Pseudo-P curve construction test: After the initial discharge, the cells were charged at 0.5 C, 1 C, or 2 C to different cutoff voltages (4.08–4.20 V) and then discharged at 0.05 C to 2.75 V. The onset of

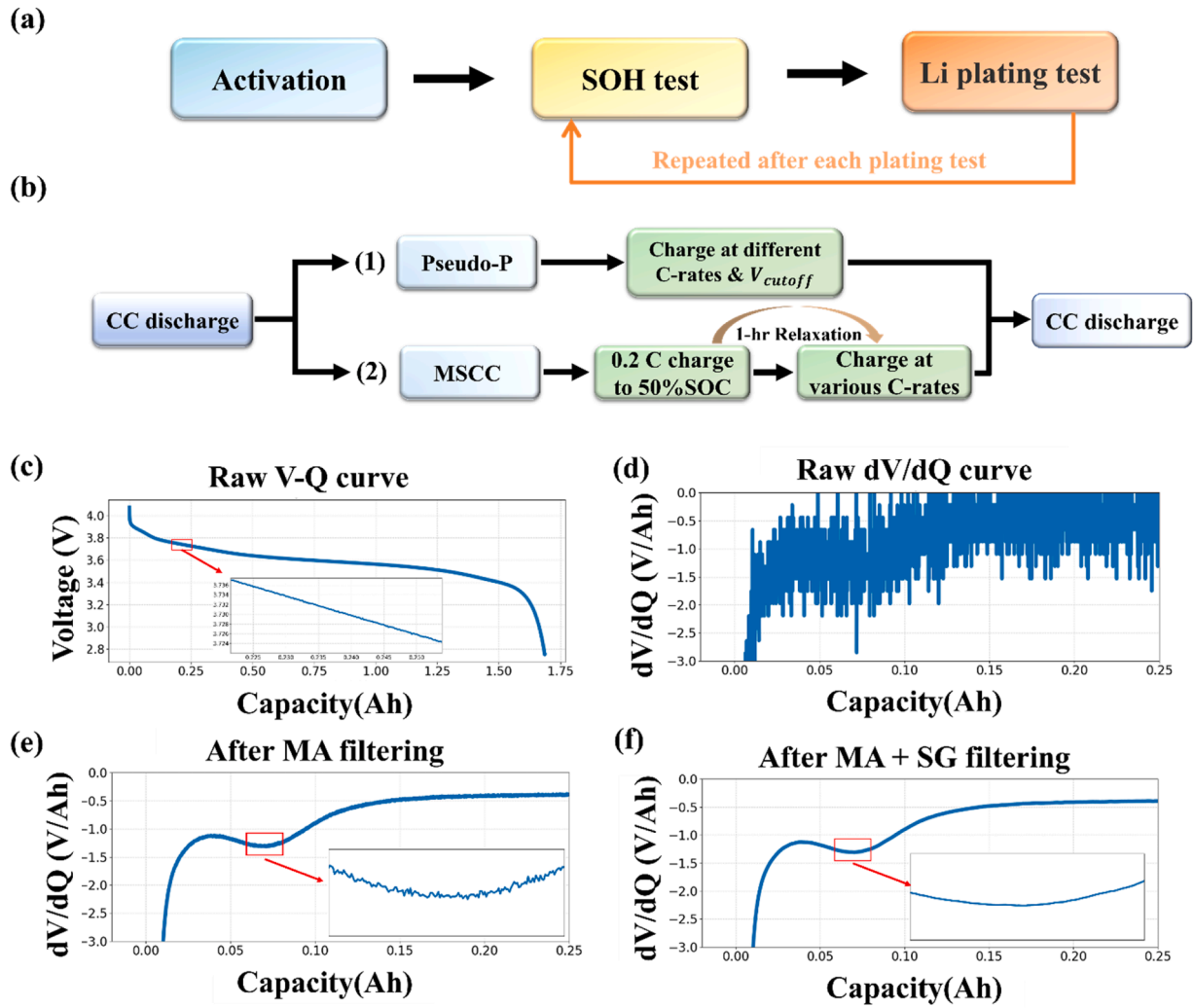


Fig. 5. (a) Schematic of the three experimental works. (b) Detailed workflow of the activation, SOH evaluation, and plating tests. (c) Raw V-Q curve with small fluctuations. (d) dV/dQ curve from the raw data. (e) After MA filtering (window: 80). (f) After MA and Savitzky-Golay filtering (order 3, window: 31).

lithium deposition was determined by performing the dV/dQ analysis on the second discharge. These results were used to establish the pseudo-P curves under different charging conditions.

(2) Strategy verification test with 50 % initial SOC: After the initial discharge, the cells were pre-charged to 50 % SOC at 0.2 C, rested for 1 h, then charged at 0.5–2 C to 4.2 V and discharged again at 0.05 C. Based on prior findings that plating initiates above 50 % SOC at high C-rates, this specific SOC was chosen to access the multi-stage constant current (MSCC) strategy, which reduces current after 50 % SOC to avoid plating.

### 3.3. Signal smoothing

The accurate identification of the valley point in the dV/dQ curve during discharge is critical, as this point signifies the end of lithium stripping and enables precise determination of the net discharge capacity  $\beta Q$ . Following established procedures that highlight the importance of signal preprocessing with smoothing filters [55], we implement a two-stage smoothing process as shown in Fig. 5(c)–(f). This process effectively reduces noise amplification when performing numerical differentiation of voltage–capacity (V–Q) experimental data.

Fig. 5(c) shows the raw V–Q curve in discharge with subtle fluctuations, where minor fluctuations lead to a noisy dV/dQ profile in Fig. 5(d), making it difficult to distinguish critical features such as the lithium stripping valley. Initially, background noise was reduced using a moving

average (MA) filter with an 80-point window as depicted in Fig. 5(e) [56]. Subsequently, a third-order Savitzky-Golay (SG) filter with a 31-point window was further applied to eliminate remaining fluctuations [57]. Fig. 5(f) displays the final outcome, showing a smooth and stable dV/dQ curve with well-preserved valley features, enabling reliable identification.

## 4. Results and discussions

### 4.1. Points on the pseudo-P curve

As shown in Figs. 3 and 4, each discharge–charge–discharge process conducted at a designated charging cutoff voltage yields a distinct set of parameters: a specific CC charging time ( $T_B$ ) signifying the transition from charging to discharging, a corresponding reversible lithium capacity ( $Q_p$ ), a specific moment ( $T_C$ ) marking the end of lithium stripping, and a quantifiable net discharge capacity ( $Q_p^*$ ). Both  $T_C$  and  $Q_p^*$  can be extracted from the dV/dQ curve, which is the first derivative of the voltage–charge curve during the second discharge process. By changing the charging cutoff voltage, the relationship between the  $Q_p^*$  and the charging time  $T_B$  can be established. Fig 6(a) displays this relationship when the charging C-rate is 1 C as well as the voltage evolution. It should be emphasized that each  $Q_p^*$  value at a given time  $T_B$ —corresponding to a specific charging cutoff voltage—is derived from the subsequent discharge phase and represents the net discharge capacity within the  $T_B$ –

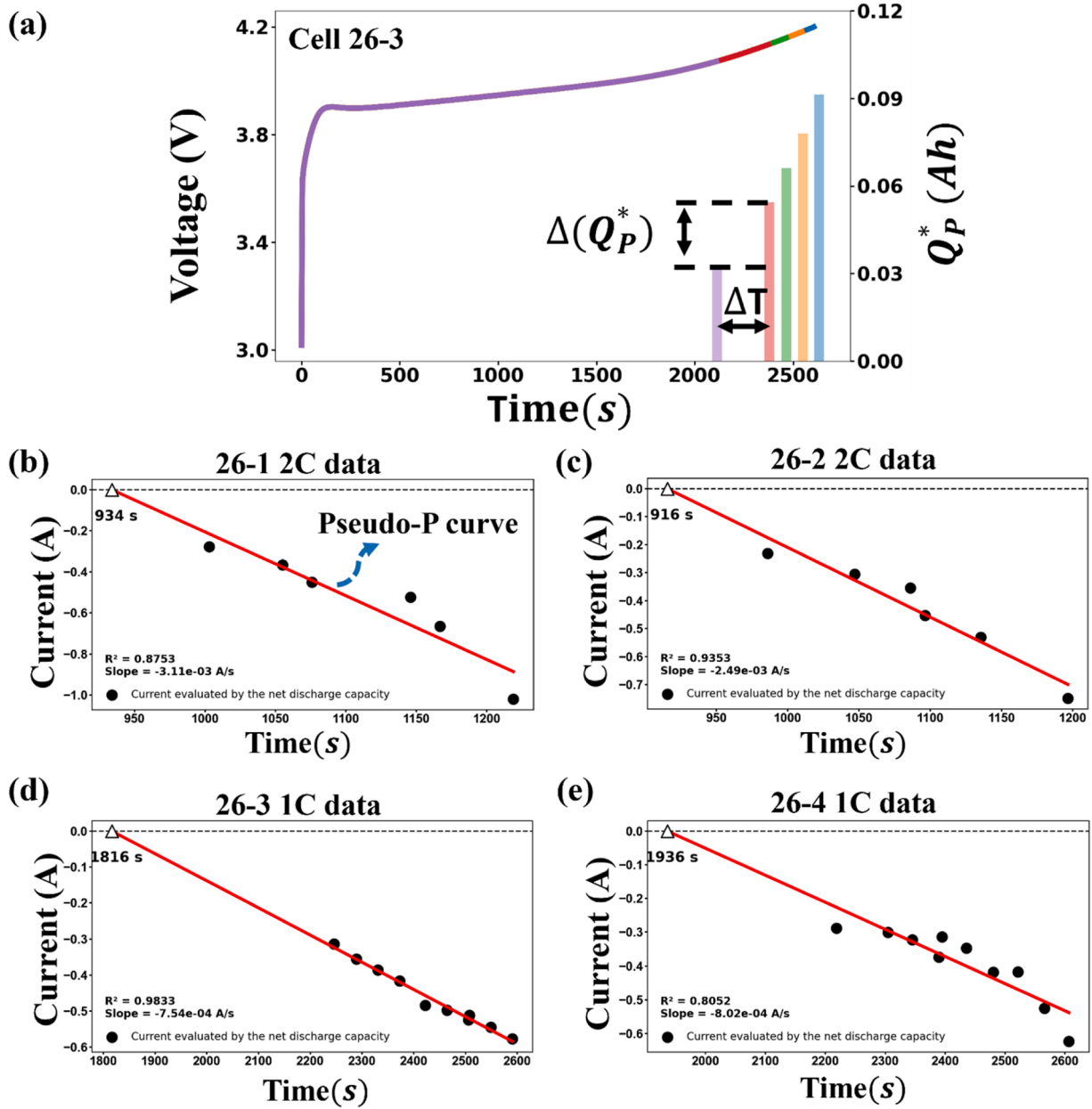


Fig. 6. (a) Voltage and  $Q_p^*$  versus  $T_B$  when the charging C-rate is 1 C. Pseudo-P curve construction at 0 °C for different cells: (b) Cell 26–1, (c) Cell 26–2, (d) Cell 26–3, and (e) Cell 26–4.

$T_C$  interval.

With these  $Q_p^*$  and their associated  $T_B$ , one can evaluate  $\Delta Q_p^* / \Delta T_B$  for two different  $T_B$  values to generate those data points that form the pseudo-P curve. As shown in our simulations (Fig. 3(b) and 4(a)) and supported by literature [58], the plating current displays near-linear behavior when the overpotential is small. Thus, fitting these data points with a straight line intersecting the baseline at the plating onset  $T_A$  is justified. As for the validity of the linearity of these points, a detailed discussion will be provided in Section 4.7. To verify repeatability under the same test conditions, four batteries (Cells 26–1 to 26–4) are tested at 0 °C, with Cells 26–1 and 26–2 charged at 2 C, and Cells 26–3 and 26–4 at 1 C. As shown in Fig. 6(b)–6(e), while there are some differences in predicting  $T_A$  when the current extrapolates to zero for individual batteries, the capacity deviation between the two batteries is only 1.00 % under 2 C charging (Cell 26–1 and 26–2) and 3.33 % under 1 C charging (Cell 26–3 and 26–4).

#### 4.2. Experimental check of $T_A$ using neighboring points

The time point  $T_A$  identified from the proposed linear pseudo-P curve is now verified to correctly indicate the onset of lithium plating, without battery disassembly. Additional experiments are conducted at 0 °C using identical charging/discharging protocols but with charging durations that deviate from the  $T_A$  value.

As illustrated in Fig. 7, selecting a 1 C charging rate and a charging duration of 1827 s results in a  $dV/dQ$  curve during the discharging phase that exhibits no signs of lithium plating, consistent with the pseudo-P curve's prediction. When the charging duration is <1827 s, no any indicators of lithium plating are observed, further confirming the absence of lithium deposition. However, once the charging time surpasses 1827 s, the  $dV/dQ$  curve begins to shift or deflect, signaling the occurrence of lithium plating during the preceding charging phase. This deviation becomes increasingly pronounced as the charging time continues to increase. A similar pattern is observed under 2 C charging conditions,



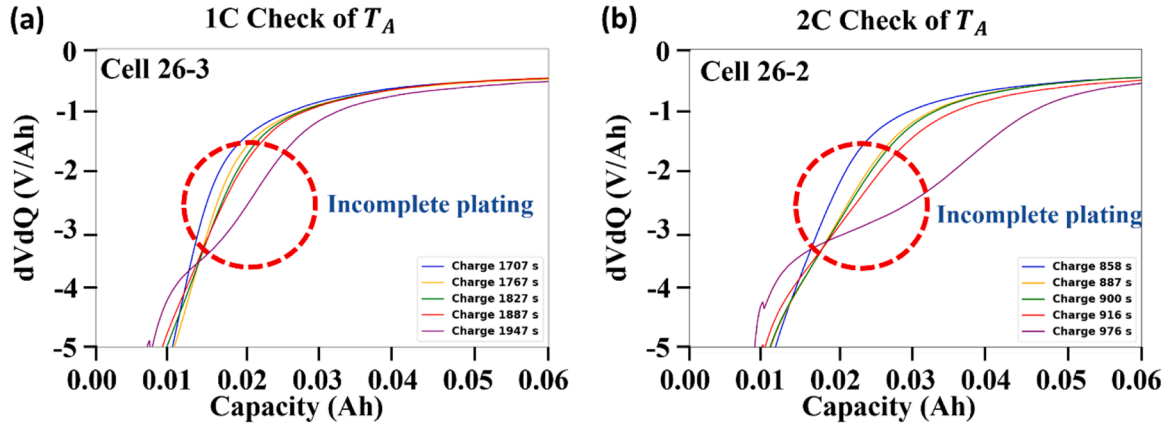


Fig. 7.  $dV/dQ$  curves in the discharging phase following charging at 0 °C with (a) 1 C-rate and (b) 2 C-rate, shown for various charging durations. Note that the experimental onset times  $T_A$  for (a) and (b) are 1816 s and 916 s, respectively.

with lithium plating initiating at 916 s (see Fig. 6(c)). Beyond this point shown in Fig. 7(b), the signs of plating become progressively more pronounced.

Fig. 7(a) demonstrates that despite  $T_A$  (=1816 s) being identified as the lithium plating onset for Cell 26–3 (see Fig. 6(d)), the  $dV/dQ$  curve in the discharging phase following this 1816-s charging duration exhibits minimal indication of plating. Even with a charging time of 1947 s (early plating stage), the  $dV/dQ$  curve displays only partial or incomplete plating characteristics [59]. This observation, also evident in Cell 26–2 with 2 C charging (see Fig. 7(b)), explains why the points used to generate the pseudo-P curve need be positioned away from the onset

point occurring at time  $T_A$  to ensure they can be clearly identified in experiment.

#### 4.3. Simulation validation

To further validate the onset of lithium plating predicted by the proposed pseudo-P curve approach accompanied with experimental full-cell data, the simulation model in PyBaMM, which is based on the P2D theory, is employed again. Although the electrochemical parameters in the simulation do not precisely match those of the tested cell, the simulation successfully reproduces key battery responses. Specifically,

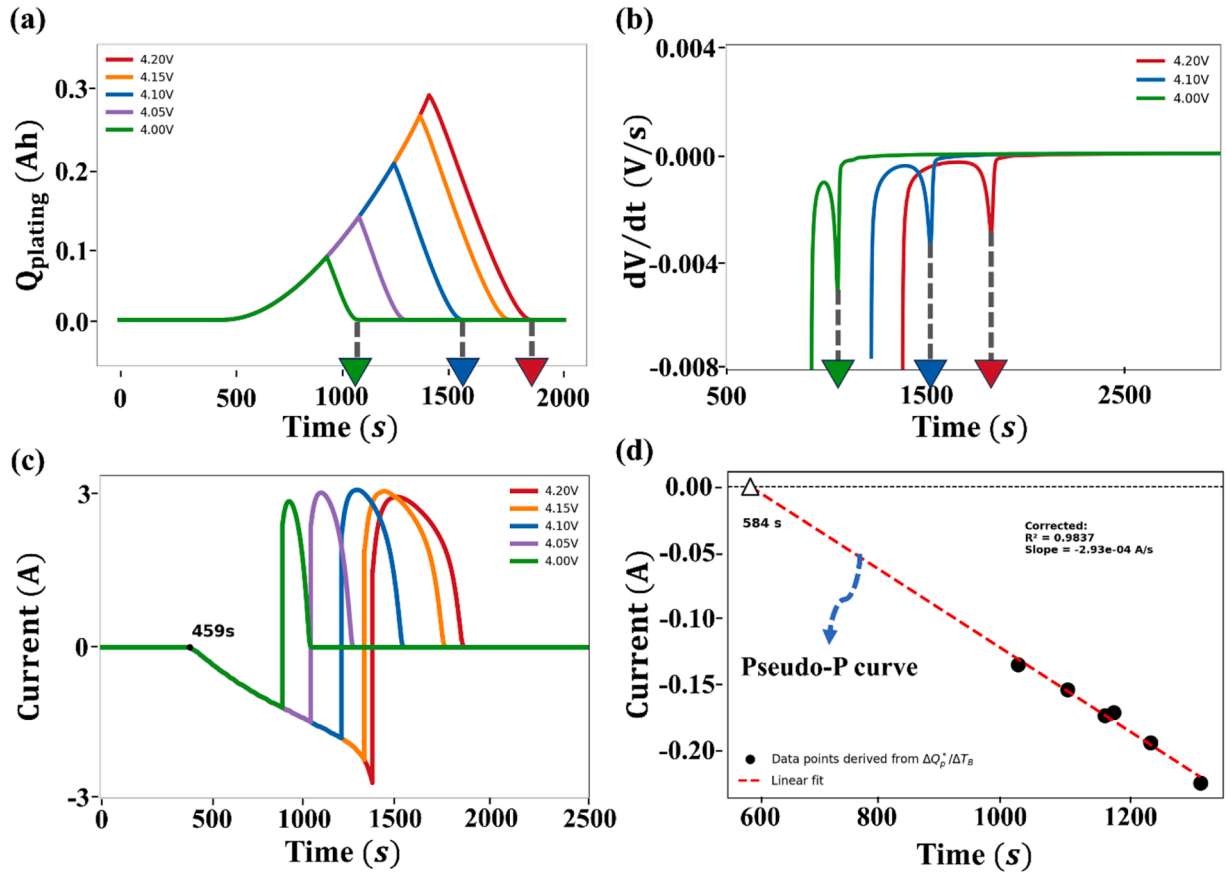


Fig. 8. (a) Simulated lithium plating deposition under 1C CC charging at 0 °C using the PyBaMM model under different charging cutoff voltages, (b)  $dV/dt$  curves during the charge/discharge process, (c) lithium plating currents, and (d) pseudo-P curve, entirely derived from the simulated data points.

the PyBaMM model provides valuable insights into mechanisms involved in challenging scenarios such as low temperatures, high currents, overcharging, and material deterioration—conditions that are dangerous and difficult to quickly reproduce in experimental settings.

These simulations estimate the lithium plating, stripping, and net charging/discharging currents by analyzing deposited/stripped lithium capacity and specific time points ( $T_B$ ,  $T_C$ ), using the methodology described in Section 4.1. All simulations are performed at 0 °C with 1 C CC charging to various cut-off voltages (4.00 V, 4.05 V, 4.10 V, 4.15 V, 4.20 V) to align with our experimental conditions.

Fig. 8(a) displays the time evolutions of plating amount under various charging cutoff voltages, with the “▼” symbol pinpointing the end of stripping ( $T_C$ ) for the three selected scenarios. These time moments also correspond to the valleys points on the respective dV/dt curves during discharging, as shown in Fig. 8(b). Since the simulations define the initial charging time as zero, the total charging duration for each case with different cutoff voltages corresponds to the value of  $T_B$ , as listed in Table 2, which also includes the maximum plating capacities. Fig. 8(c) gives the averaged plating currents across the anode thickness during the charging/discharging process, where the onset of lithium plating  $T_A$  occurs at 459 s in this simulation. Based on the simulation data for  $T_B$  and  $T_C$ , and the net discharge capacity during discharging, the coordinates of the points marked as “☆” in Fig. 4(c) can be determined. Linear fitting of this set of points produces the pseudo-P curve displayed in Fig. 8(d), yielding the simulated  $T_A$  value of 584 s. Although there appears to be a discrepancy between  $T_A$  (459 s) from the simulated plating current and that derived from the pseudo-P curve (584 s), the difference in total capacity is merely 3.44 %. This also confirms the reliability of the proposed pseudo-P curve method for estimating the onset of lithium plating.

#### 4.4. Charging with lower C-rate

From the previous experimental analysis of the two charging rates (1 C and 2 C), it is anticipated that reducing the charging rate can delay the onset of lithium plating. To examine the impact of charging rate on the onset and duration of lithium plating, additional charging data are collected to examine the scenario where the charging C-rate is reduced to 0.5 C. As depicted in Fig. 9(a), the pseudo-P curve reveals that lithium plating at 0 °C begins at 5437 s, which roughly corresponds to 75.5 % SOC, a value higher than the plating-onset SOC observed at higher charging C-rates. This can be attributed to the reduction in polarization at lower charging C-rates, enabling the battery to reach a higher SOC before the lithium plating is triggered [60].

Once we identify both the onset time of lithium plating ( $T_A$ ) from the pseudo-P curves and the moments when charging reaches the 4.2 V cutoff voltage ( $T_B$ ), we plot the lithium plating duration (the interval from  $T_A$  to  $T_B$ ) for all three charging rates (0.5 C, 1 C, and 2 C). Fig. 9(b) displays these experimental plating durations in terms of both charging time and SOC. It can be observed that as the C-rate of charging decreases, lithium plating begins later and at higher SOC.

#### 4.5. Charging starting from 50 % SOC

Fast-charging protocols typically apply high currents until a specified SOC threshold is reached, after which the current is reduced to limit

lithium deposition. To investigate this transition, we first charge the battery to an SOC of 50 % at 0.2 C, and then continue charging at varying C-rates until the cutoff voltage is reached. The SOC = 50 % is selected based on Fig. 9(b), where lithium plating occurs when the SOC exceeds 50 % for the 1 C and 2 C charging cases. This analysis enables us to assess the effects of reduced charging currents after reaching specific SOC levels and determine the safe charging duration or capacity before reaching the next plating onset point.

The pseudo-P curves for the 1 C and 0.5 C charging following an initial SOC ( $SOC_0$ ) of 50 % at 0 °C are presented in Fig. 10, with the key observations outlined below. (i) At 1 C charging rate, lithium plating initiates at 560 s, corresponding to about 15.56 % of the total capacity delivered to the battery by that time. (ii) Under 0.5 C charging, the plating onset is delayed to 1620 s, which permits approximately 22.51 % of charge capacity. These observations suggest that, at the same initial SOC, a lower C-rate enables more charge to be delivered before lithium plating begins.

#### 4.6. Charging strategy application

The proposed method allows us to detect the onset of lithium plating, enabling continued charging without plating by reducing current magnitude. To assess the performance of various charging protocols which involve the stepwise current reduction charging, we examine two variants of this charging approach: the 1 C step-charging strategy and the 2 C step-charging strategy.

Experimental data reveals that lithium deposition begins at 1816 s ( $SOC \approx 50.44$  %) when charging the battery at 1 C from 2.75 V. Reducing the charging current to 0.5 C at this point postpones plating onset. Previous analysis indicates that charging at 0.5 C from  $SOC = 50$  % requires an additional 1620 s (reaching  $SOC \approx 72.95$  %) before plating starts. The current is subsequently reduced to 0.25 C, enabling continued charging until reaching 4.2 V. A brief 5-s rest interval is introduced between current transitions to ensure smooth charging stage transitions. The procedure described above defines the 1 C step-charging strategy (1 C → 0.5 C → 0.25 C). The 2 C step-charging strategy (2 C → 1 C → 0.5 C → 0.25 C) follows a similar pattern, beginning with 2 C charging for 916 s ( $SOC \approx 50.89$  %) before switching to 1 C for an additional 560 s (reaching  $SOC \approx 66.44$  %). The current is then lowered to 0.5 C, continuing the charging process for 650 s (reaching  $SOC \approx 75.47$  %), followed by a final decrease to 0.25 C until reaching 4.2 V, where the 650 s is the time evaluated for lithium plating to occur when the battery is charged at 0.5 C from  $SOC_0$  of 65 %. This stepwise reduction in current aims to control plating by maintaining positive overpotential throughout the entire charging process [61].

For a comparative analysis between conventional CC charging and stepwise current reduction charging, Table 3 summarizes several performance metrics—including total charging time, Coulombic efficiency, total charge capacity, total discharge capacity, and net discharge capacity between  $T_B$  to  $T_C$ —across various charging protocols: traditional 2 C, 1 C, and 0.5 C methods, as well as 1 C and 2 C step-charging strategies. Note that the net discharge capacity between  $T_B$  to  $T_C$  serves as a measurable indicator for determining whether lithium plating occurred during the charging process.

First, Table 3 indicates that the step-charging approach prevents lithium plating while achieving higher charging capacity and Coulomb efficiency, though it seemingly requires a longer total charging time. Fig. 11 illustrates the validation of the absence of a stripping signal in the dV/dQ curves during the subsequent discharging process when both types of step-charging are applied. Second, in terms of total charging time, the 1 C step-charging strategy analogously functions at an average current of about 0.53 C, closely resembling the 0.5 C protocol. Thus, when comparing the conventional 0.5 C charging with the 1 C step-charging, the latter not only prevents lithium plating but also results in a shorter total charging time, a greater total charge, and improved Coulomb efficiency. Third, similarly, the 2 C step-charging functions

**Table 2**  
PyBaMM simulation data.

| Cut-off voltage (V) | Charge time (s) | Plating capacity (Ah) |
|---------------------|-----------------|-----------------------|
| 4.2                 | 1393.2          | 0.29337               |
| 4.15                | 1351.8          | 0.26628               |
| 4.10                | 1235.4          | 0.20412               |
| 4.05                | 1078.2          | 0.13444               |
| 4.00                | 933.6           | 0.08223               |

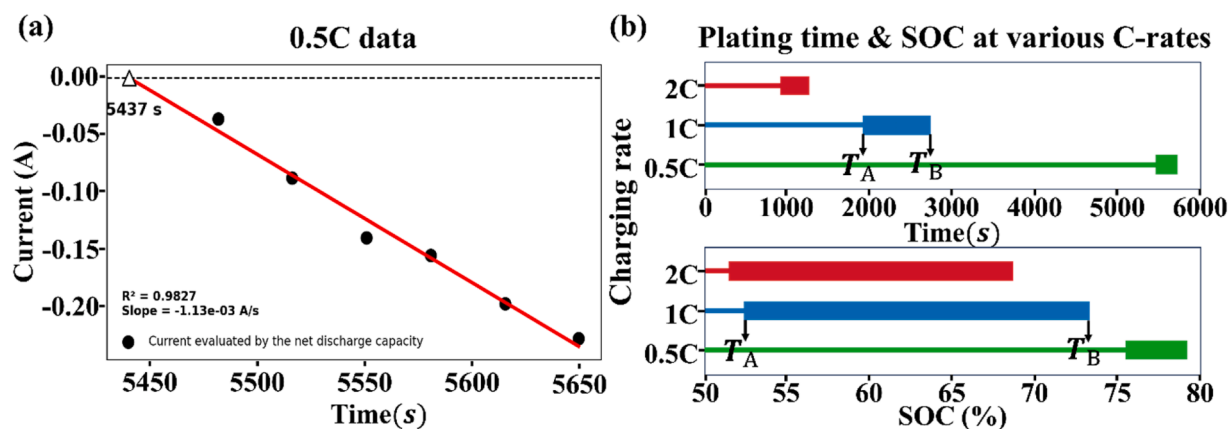


Fig. 9. (a) The pseudo-P curve generated by the experimental data points at 0 °C using 0.5 C charging current. (b) Plating durations in terms of charging time and SOC at 0 °C under three different C-rates.

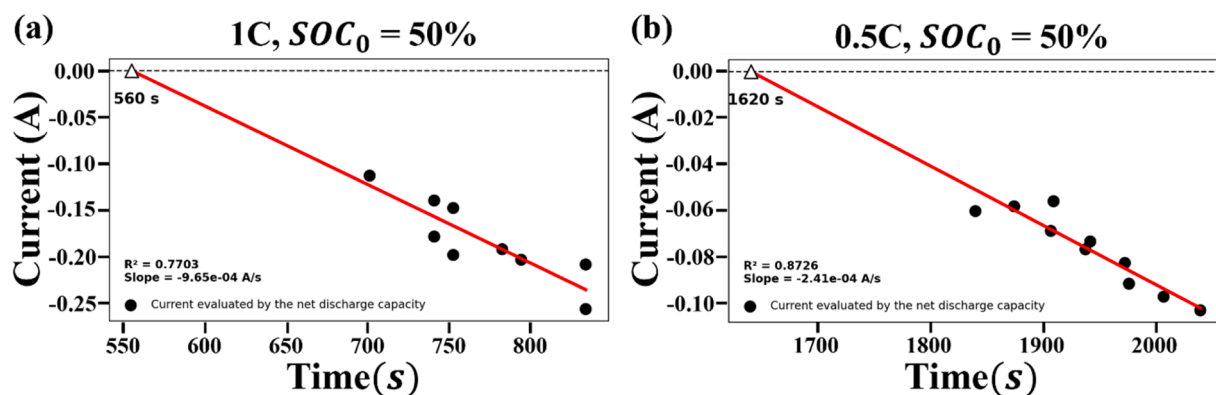


Fig. 10. Pseudo-P curves generated from data obtained when batteries with initial SOC of 50 % at 0 °C are charged at (a) 1 C and (b) 0.5 C-rates.

Table 3

Experiment data for different charging protocols.

| Charging C-rate   | Total charging time (s) | Coulomb efficiency (%) | Charge capacity (Ah) | Discharge capacity (Ah) | Net discharge from $T_B$ to $T_C$ (Ah) |
|-------------------|-------------------------|------------------------|----------------------|-------------------------|----------------------------------------|
| 2 C               | 1249                    | 93.42                  | 1.8051               | 1.6864                  | 0.0706                                 |
| 1 C               | 2605                    | 92.69                  | 1.8812               | 1.7437                  | 0.0913                                 |
| 0.5 C             | 5718                    | 94.54                  | 2.0655               | 1.9527                  | 0.0459                                 |
| 1 C step-charging | 5206                    | 95.13                  | 2.2225               | 2.1143                  | 0                                      |
| 2 C step-charging | 3570                    | 99.72                  | 2.2213               | 2.215                   | 0                                      |

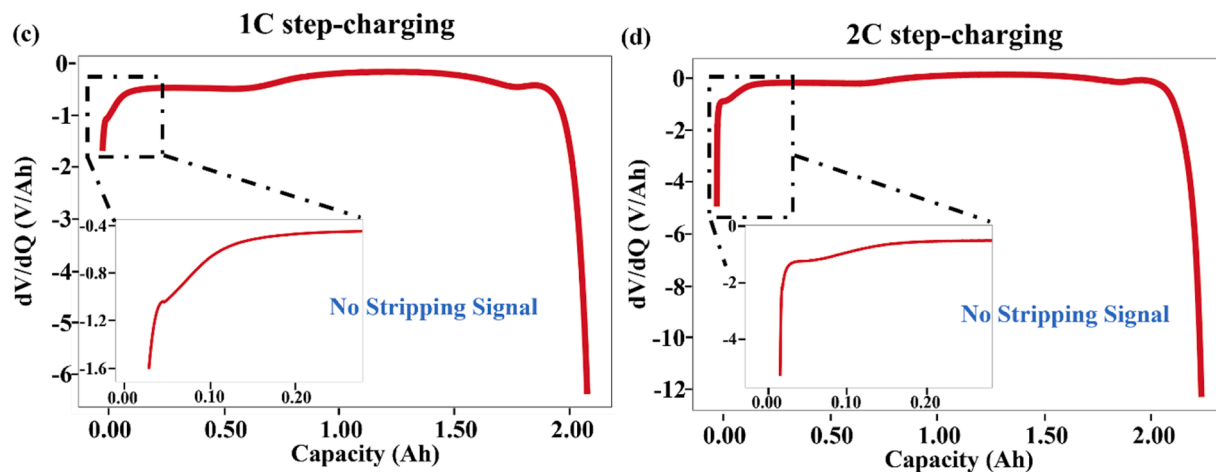


Fig. 11. The  $dV/dQ$  curves during discharge for (a) 1 C step-charging or (b) 2 C step-charging, where no stripping signal is observed.

equivalently to a conventional CC charging at 0.77 C, effectively eliminating the risk of lithium deposition while enhancing Coulombic efficiency. Fourth, among all the tested protocols, the 2 C step-charging achieves the highest Coulombic efficiency, reaching 99.72 % at 0 °C. This demonstrates its effectiveness as a charging strategy while completely avoiding lithium deposition risks.

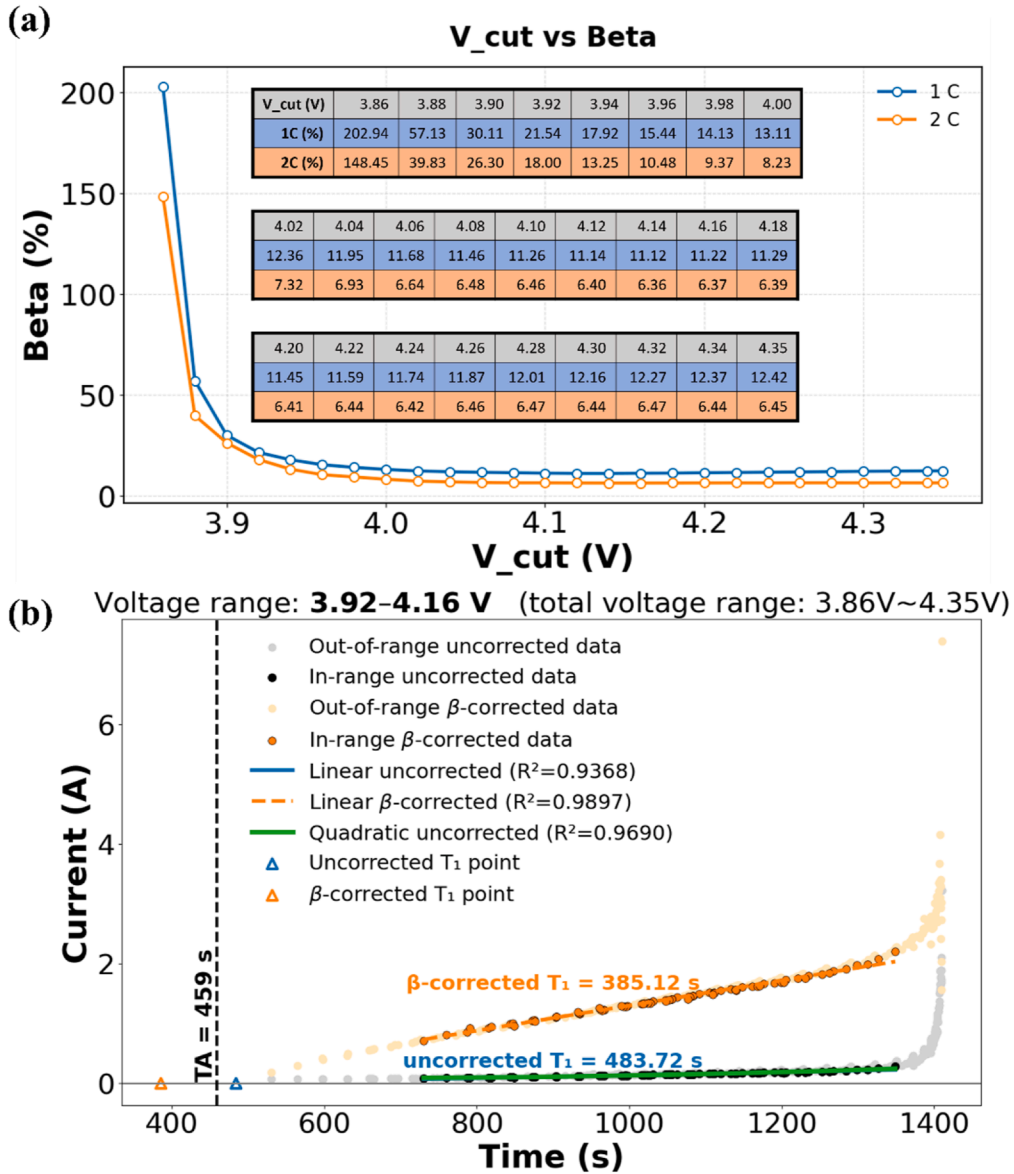
#### 4.7. Discussions

Three aspects require further clarification: (A) examination of the constant  $\beta$  ratio, (B) linear fitting of the data points, and (C) comparison with other conventional approaches.

(A)  $\beta$  ratio—This ratio is given as the net discharge capacity between  $T_B$  and  $T_C$  divided by the amount of lithium stripped during the same interval. In our simulations, the reaction rate constant of the plating reaction  $k_{pl}$  is set to  $5.5 \times 10^{-8}$  m/s [53], which influences the exchange current density for the lithium plating reaction  $j_{0,2}$  in Eq. (3). As a result, the  $\beta$  ratio is found to be approximately 11.94 % under 1 C charging across various cutoff voltages. Subsequent simulations reveal that the  $\beta$

ratio increases to around 23.30 % at 0.5 C and decreases to approximately 7.01 % at 2 C charging. A detailed simulation is further performed to evaluate the  $\beta$  ratio under various scenarios. Fig. 12(a) indicates that  $\beta$  declines rapidly and converges to a nearly constant value as the cut-off voltage increases from 3.86 V to 4.2 V, and further to 4.35 V. The specific  $\beta$  values are provided in the inset. As long as a small cut-off voltage is avoided (from 3.86 V to 4.04 V), the  $\beta$  value can be approximately regarded as constant.

However, if the detailed values  $\beta$  are taken into account, the construction of the two curves requires re-evaluation. Focusing on the 1 C-rate condition, we record the net discharge under different cut-off voltage conditions (from 3.86 V to 4.35 V) and, using the procedure outlined in Fig. 4(c), derive both the pseudo-P data points and the  $\beta$ -corrected points. To be explicit, the pseudo-P data points are derived from the evaluation of various  $\Delta Q_p^+/\Delta T_B$ , and the  $\beta$ -corrected points are obtained by multiplying a factor of  $(1/\beta)$ . Together, these datasets define the pseudo-P and P curves presented in Fig. 12(b). In this figure, Li plating begins at  $T_A = 459$  s, and several additional observations can be made. (i) Using only the cut-off voltages between 3.92 V and 4.16 V



**Fig. 12.** (a)  $\beta$  ratios determined at various cutoff voltages under 1 C-rate and 2 C-rate charging conditions. (b) Linear fitting performed separately on the original data points and their  $\beta$ -corrected counterparts over the limited voltage range from 3.92 V to 4.16 V.  $T_1$  indicates the moment at which the fitting curve meets the baseline. Quadratic fitting is also applied to the original data within the same voltage window.



yields the black data points, which form a highly linear pseudo-P curve intersecting the baseline at  $T_1 = 483$  s (5 % deviation from  $T_A$ ). (ii) After applying  $\beta$  correction to each point individually, the data corresponding to cut-off voltages between 3.92 V and 4.16 V form the almost straight orange line, which intersects the baseline at  $T_1 = 385$  s (16 % deviation). (iii) Upon consideration of these additional cut-off voltages (4.20 V to 4.35 V), the corresponding grey points deviate from the straight line in (i), with deviations most evident at the later charging stage. When the previously examined cut-off voltage range (3.92 V–4.16 V) is extended to 3.86 V–4.18 V, it can be found that  $T_1 = 403$  s (12 % deviation) for the pseudo-P curve and  $T_1 = 409$  s (11 % deviation) for the P curve. Comparing with the cut-off voltages (4.00 V to 4.20 V) adopted in the previous simulation, the pseudo-P curve leads to  $T_1 = 584$  s (27 % deviation, see Fig. 8). This substantial deviation arises from the use of datasets obtained at higher cut-off voltages, which produce a larger baseline intercept. In any case, our analysis reveals that even when slight variations in  $\beta$  result in different baseline intersection times  $T_1$  for the two curves, these intersection points remain near the target  $T_A$ .

Another simulation tests confirm that variations in the  $k_{pl}$  give rise to different  $\beta$  values; however, the fundamental conclusions outlined above remain unaffected. Even though the  $\beta$  ratio can vary with the charging rate, the pseudo-P curve reliably intersects the P curve at the lithium-plating onset time  $T_A$  as long as  $\beta$  is held constant. This same intersection validates the robustness and utility of the pseudo-P curve approach. If the experimental  $\beta$  ratio can be obtained, the total lithium plating can be estimated from the net discharge capacity, offering valuable insight for future studies on lithium deposition.

(B) Linear fitting—From the preceding analysis, the linear-fit technique employed in constructing the pseudo-P curve appears plausible within certain limited cutoff voltage ranges. To examine the linearity, we apply a quadratic fit to the same dataset (i.e., the cutoff-voltage data from 3.92 V to 4.16 V) to evaluate whether a higher-order fit offers improvement over the linear fit with  $R^2 = 0.9368$ . It is observed that although the quadratic fit yields a slightly higher  $R^2$  value of 0.9690, the resulting curve (green in Fig. 12(b)), however, fails to intersect the baseline. The absence of an intersection with the baseline may be attributed to the inclusion of right-side data points that exhibit slight upward bias, which forces the second-order fit upward. This finding suggests that the quadratic fit may be ineffective for estimating the Li-plating onset. Therefore, instead of pursuing higher-order fits for marginally better agreement, the priority should be selecting a proper cutoff-voltage range to obtain a reliable linear pseudo-P curve.

(C) Comparison with other methods—As shown in Table 1, several non-invasive methods have been employed to examine lithium plating. To highlight the distinguishing features of our proposed method, we compare it with four conventional techniques: CE (Coulombic efficiency), IC (incremental capacity), DV (differential voltage), and EIS (electrochemical impedance spectrum).

First, CE is highly sensitive to lithium metal side reactions, but it measures the total irreversible capacity across an entire cycle rather than offering time-dependent data [62]. Additionally, CE is affected by side reactions like SEI growth and lithium migration or redistribution after rest, making it difficult to distinguish lithium plating from other electrochemical phenomena. When considering only reversible lithium plating, CE remains at 100 % even if charging extends beyond the lithium plating onset point; therefore, CE cannot indicate when lithium plating begins.

Second, the IC signal appears only after lithium plating has caused noticeable distortion in the voltage–capacity curve. In other words, IC may not be able detect the true onset of plating but rather indicates the point at which accumulated plating begins to measurably affect the curve [19]. To confirm this, IC analysis is conducted on data from charging at 0 °C and 1 C to 4.2 V. The dQ/dV curve displays two peaks at 3.8872 V and 4.0676 V in our simulation, representing the so-called intercalation and plating peaks, both after the plating onset at 3.8467 V (start of plating current). This shows that the IC peak is not an accurate

indicator to pinpoint the onset of plating.

Third, DV acts as a posterior signal, detecting plating only after it has noticeably affected the voltage, rather than at the exact moment plating initially begins [59]. Indeed, one element of our approach developed in this study involves using DV curves to determine when lithium stripping ends during discharge. Relying solely on DV curves is insufficient for determining the correct plating onset because lithium stripping must reach a sufficient level before it becomes detectable in the voltage response. In other words, accurate measurement of the stripping amount requires the charging cutoff voltage to exceed the plating onset voltage by a substantial margin.

Fourth, EIS plating detection relies on impedance changes: decreased 1 Hz impedance after sufficient lithium deposition or semicircle contraction of dynamic EIS (DEIS) once plating reaches detectable levels [18,27]. These changes inherently require that interfacial impedance has already been altered by metallic lithium, and therefore an unavoidable detection delay exists. In essence, EIS/DEIS methods detect when impedance changes reach measurable levels, whereas the proposed pseudo-P approach is able to identify the actual onset of lithium plating.

Unlike conventional techniques, our method yields time-dependent plating current profiles by integrating DV curves collected under various cutoff conditions with measured net discharge capacity. Its advantages are as follows. (i) The plating onset is obtained, without relying on voltage distortion—a common limitation of IC. (ii) It captures the moment when pure intercalation gives way to concurrent intercalation and plating, a transition marked by zero overpotential and having a clear physical interpretation. (iii) It reveals lithium-plating kinetics and informs the design of follow-up charging strategies. In summary, the temporal resolution and dynamic characterization provided by this method are unavailable in CE, IC, DV, and EIS approaches, which cannot reliably pinpoint plating onset.

## 5. Conclusion

Monitoring lithium plating in commercial LIBs without disassembly is essential for risk management and safety evaluation during operation. This study introduces an experimental pseudo-P curve approach to estimate the onset of lithium plating, relying solely on the signal acquired from a full-cell measurement. Based on the electrochemical simulation outcomes showing a consistent  $\beta$  ratio between the net discharge amount and the lithium stripping during discharging (equivalent to the lithium plating during charging when only reversible plating is taken into account) at the same C-rate, we construct a pseudo-P curve that intersects with the plating current curve (P curve). This intersection point marks the precise moment when lithium plating begins. The construction process requires: (i) measuring the net discharge capacity experimentally by identifying the valley point on the dV/dQ discharge curve, and (ii) systematically varying the charging cutoff voltages to obtain multiple data points for linear fitting of the pseudo-P curve. The onset of lithium plating can still be predicted without requiring the exact value of the  $\beta$  ratio.

The PyBaMM simulation confirms the feasibility of the proposed pseudo-P curve approach. Moreover, its validity is also experimentally verified—without opening or compromising the battery's structural integrity—by detecting the presence or absence of stripping signals during discharging in cases where charging duration is either below or above the onset of lithium plating. Based on the identified onset moments across various charging C-rates, two step-charging protocols that completely avoid lithium plating are developed, with the 2 C step-charging protocol showing superior performance by achieving a Coulombic efficiency of 99.72 % at 0 °C.

This study is still limited to the scenarios of fresh batteries at 0 °C. As LIBs age, it is expected that, under a fixed large charging C-rate, lithium plating of aged LIBs will occur earlier than that of fresh LIBs. This is attributed to the fact that increased impedance (characteristic of aged



batteries) will inhibit lithium intercalation into the negative electrode. Lower environmental temperatures slow down reaction kinetics, also hindering lithium-ion transport and promoting the onset of lithium plating. A more comprehensive study on these topics, particularly gathering additional data to identify the onset points, will be essential for the future development of optimal charging strategies.

### CRedit authorship contribution statement

**Ying Kung:** Validation, Software, Methodology, Formal analysis.  
**Yu-Hong Zhang:** Visualization, Investigation, Data curation.  
**Kuo-Ching Chen:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Formal analysis, Conceptualization.  
**Chih-Hung Chen:** Methodology, Investigation, Conceptualization.  
**Chi-Jyun Ko:** Investigation, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Data availability

Data will be made available on request.

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